

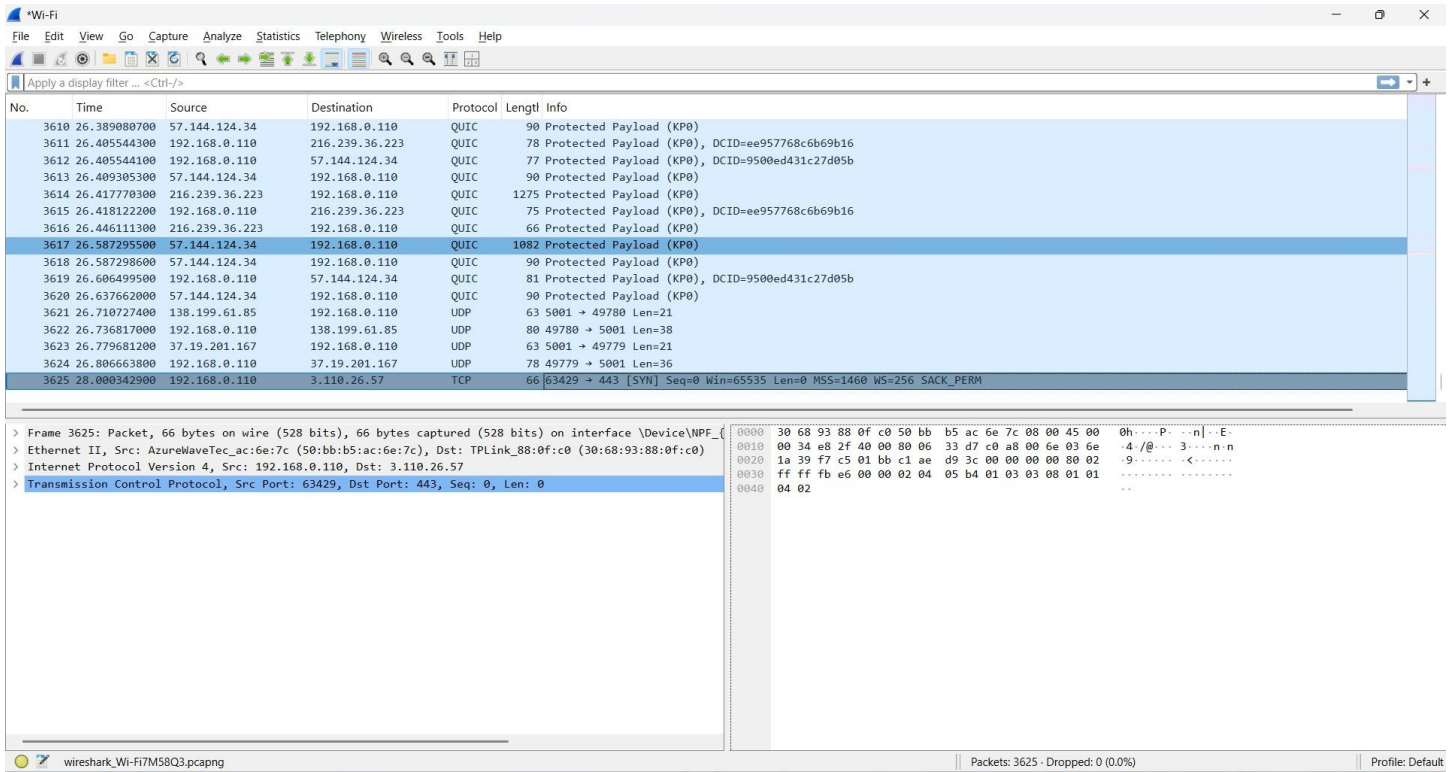
N+- Network Fundamentals and building network

[Q] Introduction to Networking and the OSI Model Theory (1 hour): What is Networking? Overview of the OSI Model: Layers and their functions. Practical (1 hour): Use a network sniffer (e.g., Wireshark) to observe traffic and identify layers. Task: Create a diagram of the OSI Model with examples for each layer.

[Q-1] Install Wireshark on your laptop, start a capture on your main network interface, and visit any website (like instagram.com). Stop the capture and note down the protocol names you see in the packet list.

[ANS]

No.	Protocol	Purpose
1	DNS	Converts website names into IP addresses
2	TCP	Reliable data transport
3	TLSv1.3	Secure encrypted communication
4	HTTP/HTTPS	Website communication
5	ARP	Finds MAC addresses on local network
6	ICMP	Network diagnostics
7	UDP	Fast communication protocol



Wireshark capture showing network traffic. The packet list shows several QUIC and UDP packets. Packet 3625 is selected, showing details for Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol. The packet bytes pane shows the raw data in hexadecimal and ASCII.

[Q-2] Pick any 3 packets from your Wireshark capture and identify which OSI layer each packet mainly represents (for example, HTTP for Application layer, TCP for Transport layer, etc.). Hint: Use the Protocol column and Info details in Wireshark to help you map protocols to OSI layers.

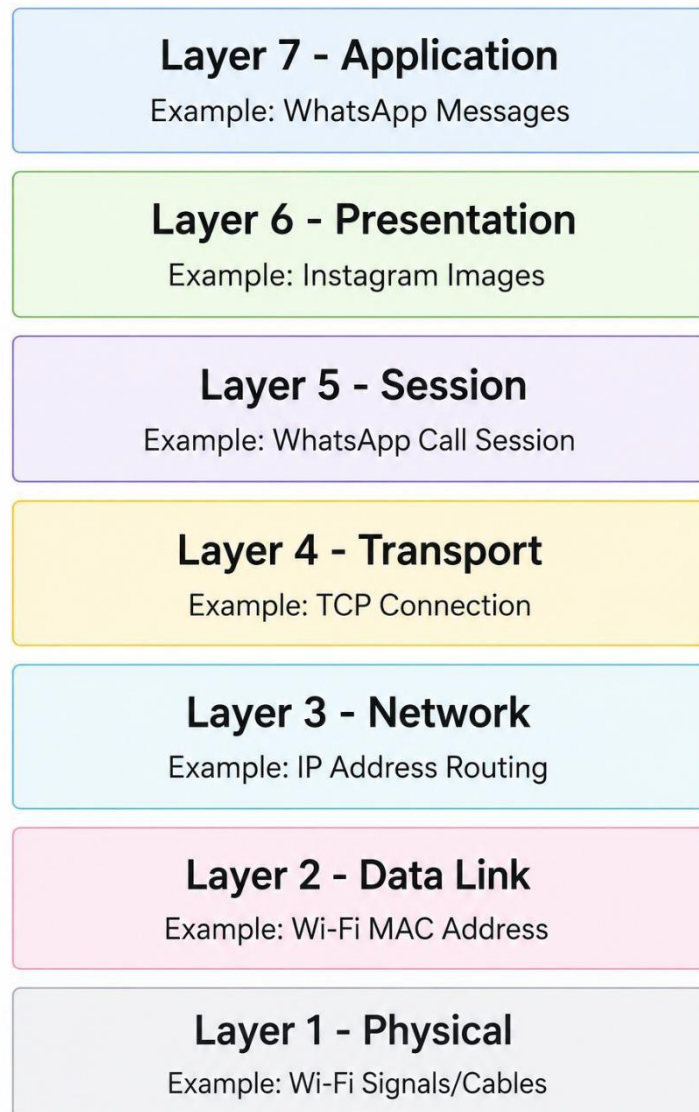
[ANS]

Packet	Protocol	Main OSI Layer	Explanation
Packet 1	DNS	Application Layer (Layer 7)	Resolves instagram.com into an IP address

Packet 2	TCP	Transport Layer (Layer 4)	Establishes reliable connection between devices
Packet 3	ARP	Data Link Layer (Layer 2)	Finds the MAC address of a device on the network

[Q-3] Create a colourful diagram of the OSI Model using any drawing tool (MS Paint, Canva, or even paper and upload a photo). For each layer, add a real-world example from apps you use daily (e.g., WhatsApp uses the Application layer, Instagram images use the Presentation layer, etc.).

[ANS]



[Q-4] Explain how sending a message on WhatsApp travels through at least 4 layers of the OSI Model, describing what happens at each layer in your own words. Hint: Think about how your message is typed, packaged, sent over the internet, and received.

[ANS]

○ Explanation

When I send a message on WhatsApp, it travels through several layers of the OSI Model.

1. Application Layer (Layer 7)

I type a message such as "Hello". WhatsApp creates the message and prepares it to be sent.

2. Presentation Layer (Layer 6)

The message is encrypted so that only the intended recipient can read it. This provides security and privacy.

3. Session Layer (Layer 5)

WhatsApp maintains a communication session between my phone and the recipient's device, ensuring both devices stay connected.

4. Transport Layer (Layer 4)

TCP divides the message into smaller segments and ensures they are delivered reliably and in the correct order.

5. Network Layer (Layer 3)

IP addresses are added, and routers determine the best path for the message across the Internet.

6. Data Link Layer (Layer 2)

The message is packaged into frames with MAC addresses so it can travel within the local network.

7. Physical Layer (Layer 1)

The data is converted into electrical signals or Wi-Fi radio waves and transmitted through the network.

Conclusion

The recipient's phone receives the signals and the process happens in reverse until the WhatsApp message appears on their screen.